

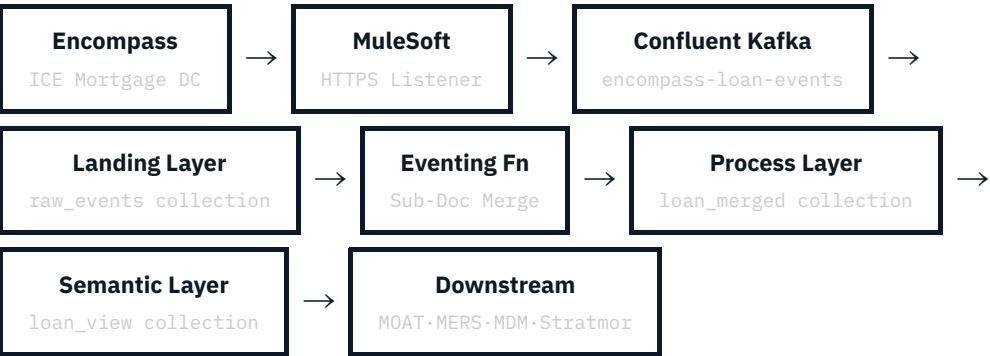
Encompass → Couchbase Integration

Design

DataHub Platform · Wintrust Financial · Initial Architecture
Draft · April 2026

DRAFT
v0.1

END-TO-END DATA FLOW



SUB-DOCUMENT API – FIELD-LEVEL UPDATE STRATEGY

WEBHOOK PAYLOAD (INPUT)

```
{
  "loanGuid":
    "LOAN-00123456",
  "fieldId":
    "CX.LOAN.AMOUNT",
  "oldValue":
    "360000",
  "newValue":
    "375000",
  "eventTimestamp":
    "2026-04-15T14:22Z"
}
```

EVENTING FUNCTION LOGIC

```
1. Map fieldId
   → doc path
   CX.LOAN.AMOUNT
   →
   "loan.amount"

2. Check loan
   doc exists?
   YES → Sub-
   Doc mutate
   NO →
   Create
   skeleton
   +
   Sub-Doc mutate

3. Atomic
   field update
   only

4. Increment
   version
```

SUB-DOC MUTATE (RESULT)

```
mutateIn("loan::LOAN-00123456",[
  upsert("loan.amount",
    375000),
  upsert("timestamps.lastModified",
    "..."),
  counter("metadata.version", +1)
])
// Only 3 fields
// touched
// Rest of doc
// untouched
```

COUCHBASE BUCKET DESIGN – MORTGAGE BUCKET

LAYER	COLLECTION	DOCUMENT KEY	PURPOSE	TTL
LANDING	raw_events	evt:: {loanGuid}:: {eventId}	Immutable audit trail of every webhook event. Replay source if reprocessing needed.	30 days
PROCESS	loan_merged	loan:: {loanGuid}	Fully merged current-state loan document. Sub-Doc updated on every field change. One doc per loan.	None (active loans)
SEMANTIC	loan_view	view:: {loanGuid}	Downstream-ready projection. Sensitive fields stripped. Pre-shaped for MOAT, MERS, MDM, Stratmor APIs.	None

KEY DESIGN DECISIONS

SAME CLUSTER – NEW BUCKET

Extend existing PRDCCDMAZCEN_001 with a dedicated **mortgage** bucket. No new cluster cost. Full logical isolation via scopes and collections.

SUB-DOCUMENT API FOR CONCURRENCY

Webhook sends only changed field. Sub-Doc mutate updates that field atomically. Five concurrent changes to same loan all succeed — no overwrite risk.

TWO EVENTING FUNCTIONS

Fn 1: landing → process (field merge + Sub-Doc upsert)

Fn 2: process → semantic (projection + sensitive field strip)

OPERATIONAL VS HISTORICAL SPLIT

Couchbase: Current state, fast N1QL, real-time queries for downstream apps.

SQL Server: Full field change history, regulatory retention, reporting.

